

EE3621: Digital Signal Processing

Student Lecture Notes — Lecture 26 Equalization, Noise Cancellation & Adaptive FIR Filters

III B.Tech EEE — Core Exam Preparation Guide

1 Channel Equalization & The Zero-Forcing Equalizer

1.1 Inter-Symbol Interference (ISI)

When digital symbols $s[n]$ are transmitted across a dispersive communication channel $h_c[n]$, multipath delays spread out adjacent symbols, creating **Inter-Symbol Interference (ISI)**:

$$x[n] = s[n] \otimes h_c[n] + v[n] = h_c[0]s[n] + \sum_{k \neq 0} h_c[k]s[n-k] + v[n] \quad (1)$$

1.2 The Zero-Forcing Equalizer (ZFE) Criterion

An equalizing filter $h_{eq}[n]$ inverts channel distortion so that the combined response is a pure delay of Δ samples:

$$H_{eq}(z) \cdot H_c(z) = z^{-\Delta} \iff \mathbf{q}[n] = \mathbf{h}_c[n] \otimes \mathbf{h}_{eq}[n] = \begin{cases} 1, & n = \Delta \\ 0, & n \neq \Delta \end{cases} \quad (2)$$

For an N -tap FIR equalizer $h_{eq}[n] = [w_0, w_1, \dots, w_{N-1}]^T$, we force $q[n] = 0$ for $n \neq \Delta$ and $q[\Delta] = 1$, setting up a system of linear algebraic equations to solve for weights w_k .

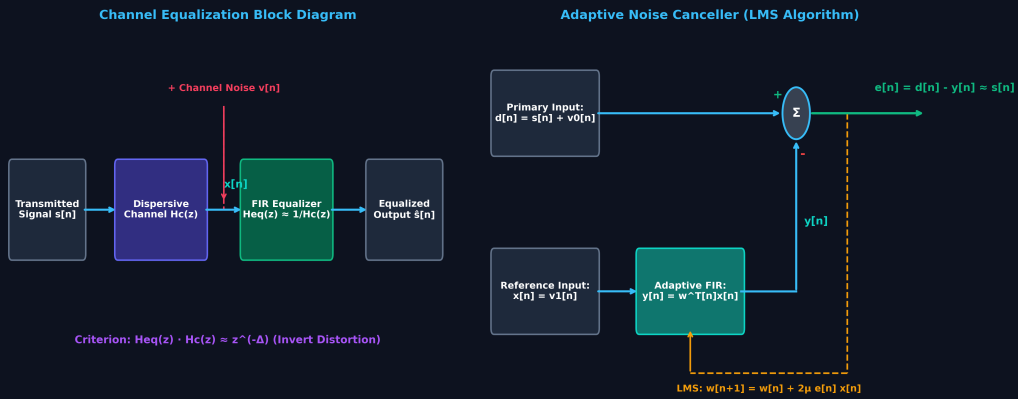


Figure 1: Left: Channel Equalizer inverting dispersive channel $H_c(z)$. Right: Dual-input Adaptive Noise Canceller driven by the LMS algorithm.

2 Adaptive Noise Cancellation (ANC)

2.1 Dual-Input Architecture

Unlike conventional static filtering, Adaptive Noise Cancellation uses two separate inputs:

- **Primary Input** $d[n]$: Desired signal $s[n]$ contaminated by noise $v_0[n]$:

$$d[n] = s[n] + v_0[n] \quad (3)$$

- **Reference Input** $x[n]$: Noise $v_1[n]$ that is correlated with $v_0[n]$, but **completely uncorrelated with the signal** $s[n]$ ($E[s[n] \cdot v_1[n - k]] = 0$).

2.2 Mathematical Proof of Signal Extraction

The system error output is the difference between primary input and filter estimate $y[n]$:

$$e[n] = d[n] - y[n] = s[n] + (v_0[n] - y[n]) \quad (4)$$

Squaring and taking the statistical expectation:

$$E[e^2[n]] = E[s^2[n]] + E[(v_0[n] - y[n])^2] + 2E[s[n](v_0[n] - y[n])] \quad (5)$$

Since $s[n]$ is uncorrelated with both $v_0[n]$ and $y[n]$, the cross-term is zero:

$$\mathbf{E}[e^2[\mathbf{n}]] = \mathbf{E}[s^2[\mathbf{n}]] + \mathbf{E}[(\mathbf{v}_0[\mathbf{n}] - \mathbf{y}[\mathbf{n}])^2] \quad (6)$$

The Core Principle: Signal power $E[s^2[n]]$ is unaffected as filter weights adapt. Therefore, minimizing total output power $E[e^2[n]]$ forces:

$$y[n] \longrightarrow v_0[n] \implies \mathbf{e}[\mathbf{n}] \longrightarrow \mathbf{s}[\mathbf{n}] \quad (7)$$

The system error output $e[n]$ becomes the ****clean, noise-free signal****!

3 The Adaptive FIR Filter & The LMS Algorithm

3.1 The 3 Core Operational Equations of LMS

For an M -tap adaptive transversal FIR filter with weight vector $\mathbf{w}[n] = [w_0[n], \dots, w_{M-1}[n]]^T$ and input vector $\mathbf{x}[n] = [x[n], \dots, x[n - M + 1]]^T$:

1. **Filter Output:**

$$\mathbf{y}[\mathbf{n}] = \mathbf{w}^T[\mathbf{n}]\mathbf{x}[\mathbf{n}] = \sum_{k=0}^{M-1} \mathbf{w}_k[\mathbf{n}]\mathbf{x}[\mathbf{n} - \mathbf{k}] \quad (8)$$

2. **Error Estimation:**

$$\mathbf{e}[\mathbf{n}] = \mathbf{d}[\mathbf{n}] - \mathbf{y}[\mathbf{n}] \quad (9)$$

3. **Widrow–Hoff Weight Update:**

$$\mathbf{w}[\mathbf{n} + 1] = \mathbf{w}[\mathbf{n}] + 2\mu \mathbf{e}[\mathbf{n}] \mathbf{x}[\mathbf{n}] \quad (10)$$

or for each individual tap: $\mathbf{w}_k[\mathbf{n} + 1] = \mathbf{w}_k[\mathbf{n}] + 2\mu \mathbf{e}[\mathbf{n}] \mathbf{x}[\mathbf{n} - \mathbf{k}]$.

3.2 Step-Size Parameter μ Stability Criterion

To guarantee convergence without divergent numerical instability:

$$0 < \mu < \frac{1}{M \cdot P_x} \quad (11)$$

where M is the number of filter taps and $P_x = E[x^2[n]]$ is the input signal power.

4 Standard Solved Exam Example

Problem: A 2-tap adaptive FIR filter has initial weights $\mathbf{w}[0] = [0.0, 0.0]^T$ and adaptation step-size $\mu = 0.1$. Given the input sequences:

- At $n = 0$: Reference vector $\mathbf{x}[0] = [1.0, 0.5]^T$, Primary target $d[0] = 2.0$.
- At $n = 1$: Reference vector $\mathbf{x}[1] = [0.5, 1.0]^T$, Primary target $d[1] = 1.5$.

Compute the filter output $y[n]$, error $e[n]$, and updated weight vector $\mathbf{w}[n+1]$ for iterations $n = 0$ and $n = 1$.

Solution:

Iteration 1 ($n = 0$):

1. **Filter Output $y[0]$:**

$$y[0] = w_0[0]x[0] + w_1[0]x[-1] = (0.0)(1.0) + (0.0)(0.5) = \mathbf{0.0} \quad (12)$$

2. **Error $e[0]$:**

$$e[0] = d[0] - y[0] = 2.0 - 0.0 = \mathbf{2.0} \quad (13)$$

3. **Weight Vector Update $\mathbf{w}[1]$:** With $2\mu = 2(0.1) = 0.2$:

$$2\mu e[0] = 0.2 \times 2.0 = 0.4 \quad (14)$$

$$w_0[1] = w_0[0] + 0.4x[0] = 0.0 + 0.4(1.0) = \mathbf{0.40} \quad (15)$$

$$w_1[1] = w_1[0] + 0.4x[-1] = 0.0 + 0.4(0.5) = \mathbf{0.20} \quad (16)$$

Updated weight vector: $\mathbf{w}[1] = [0.40, 0.20]^T$.

Iteration 2 ($n = 1$):

1. **Filter Output $y[1]$:**

$$y[1] = w_0[1]x[1] + w_1[1]x[0] = (0.40)(0.5) + (0.20)(1.0) = 0.20 + 0.20 = \mathbf{0.40} \quad (17)$$

2. **Error $e[1]$:**

$$e[1] = d[1] - y[1] = 1.5 - 0.40 = \mathbf{1.10} \quad (\text{Error reduced from 2.0 to 1.10!}) \quad (18)$$

3. **Weight Vector Update $\mathbf{w}[2]$:**

$$2\mu e[1] = 0.2 \times 1.10 = 0.22 \quad (19)$$

$$w_0[2] = w_0[1] + 0.22x[1] = 0.40 + 0.22(0.5) = 0.40 + 0.11 = \mathbf{0.51} \quad (20)$$

$$w_1[2] = w_1[1] + 0.22x[0] = 0.20 + 0.22(1.0) = 0.20 + 0.22 = \mathbf{0.42} \quad (21)$$

Final weight vector: $\mathbf{w}[2] = [0.51, 0.42]^T$.

5 Essential Exam Checklist & Traps

- **LMS Order of Operations:** Always compute in strict order: $y[n] \rightarrow e[n] \rightarrow \mathbf{w}[n+1]$. Never update weights before computing the output!
- **The Factor of 2 in LMS:** Remember the update is $\mathbf{w}[n+1] = \mathbf{w}[n] + 2\mu e[n]\mathbf{x}[n]$.
- **Reference Input Isolation:** In ANC, the reference input $x[n]$ must contain **only** noise; it must never contain the desired signal $s[n]$.
- **Step-Size Bounds:** Selecting $\mu > 1/(MP_x)$ leads to immediate mathematical divergence ($w_k \rightarrow \infty$).